



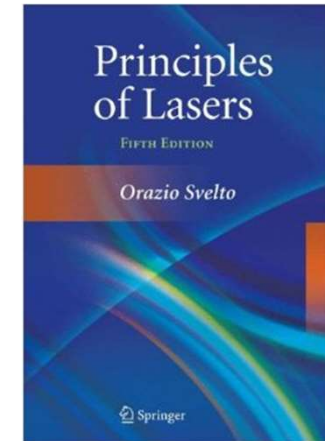
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch5-1

Characteristics of Laser Amplification



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Chapter 5: Amplification Characteristics

❖ 5.1 Classification of Laser Amplifiers

- Classification by time
- Classification by working mode
- Classification by functionality and optical excitation

❖ 5.2 Gain Characteristics of the Uniform Excited Continuous Laser Amplifier

- Gain of the laser amplifier for weak signal
- Gain of the laser amplifier for strong signal (saturated)
- Maximum output intensity
- Width of amplifier gain spectrum and output line profile

❖ 5.3 Gain Characteristics of the Longitudinal Excited Continuous Laser Amplifier

5.1 Classification of Laser Amplifiers

❖ Classification by time

Relaxation process: establishment or extinction of a certain state (relaxation time)

- Longitudinal relaxation T_1 : change of population at the energy level
- Transverse relaxation T_2 : time of generation and extinction of macroscopic polarized induction
 - EM resonant interaction: in phase, $\Delta \nu_L$ dominates $T_2 = \tau_L = 1/\pi \Delta \nu_L$
 - Collision and other process: out of phase, $\Delta \nu_D$ dominates $T_2^* = 1/2 \Delta \nu_D$

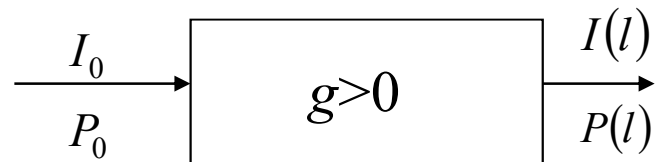
According to the incident pulse width τ_0 and relaxation time T of gain material

- | | | |
|------------------------------------|------------------------|------------------------|
| ✓ Continuous laser amplifier | $T_1 < \tau_0$ | Steady-state method |
| ✓ Pulse laser amplifier | $T_2 \ll \tau_0 < T_1$ | Nonsteady-state method |
| ✓ Ultrashort pulse laser amplifier | $\tau_0 < T_2$ | Semi-classical theory |

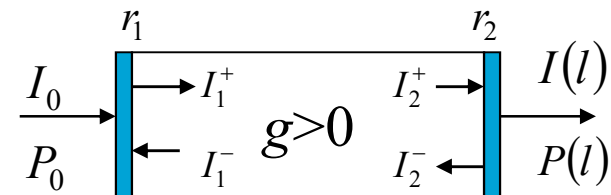
Time (s)	Solid	Gas	Semiconductor
Longitudinal relaxation time	$10^{-3} \sim 10^{-4}$	$10^{-6} \sim 10^{-9}$	10^{-9}
Transverse relaxation time	$10^{-11} \sim 10^{-12}$	$10^{-8} \sim 10^{-9}$	10^{-12}

❖ Classification by working mode

Traveling wave amplifier



Regenerative amplifier (F-P amplifier)

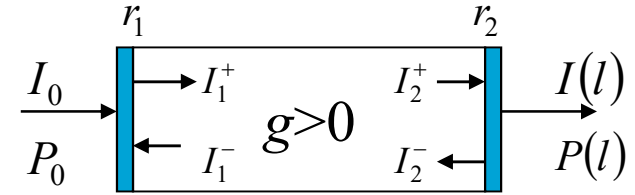


- Traveling wave amplifier: the incident light frequency is in the range of the gain spectrum.
- Regenerative amplifier: the incident light frequency needs to be close to the resonant frequency of the resonator to guarantee frequency matching.

Gain of the amplifier $G = \frac{I(l)}{I_0} = \frac{P(l)}{P_0}$ $G_{dB} = 10 \lg G$

Gain of regenerative amplifier

Multi-beam interference treatment



$$G_s = \frac{I_2^+}{I_1^+} = \frac{I_1^-}{I_2^-} \Rightarrow G = \frac{(1-r_1)(1-r_2)G_s}{\left(1 - \sqrt{r_1 r_2} G_s\right)^2 + 4\sqrt{r_1 r_2} G_s \sin^2 \left[\frac{2\pi l}{\nu} (\nu - \nu_c) \right]}$$

When $\nu = \nu_c$ $G = \frac{(1-r_1)(1-r_2)G_s}{\left(1 - \sqrt{r_1 r_2} G_s\right)^2}$ maximum gain

- ν deviates from $\nu_c \rightarrow G \downarrow$
- The larger r_1 or r_2 , the allowed bigger deviation of ν from $\nu_c \rightarrow G \uparrow$

Example of regenerative amplifier: semiconductor amplifier $\eta=3.2-3.4$

$$\Delta G = \frac{G_{FP}^{\max}}{G_{FP}^{\min}} = \left(\frac{1 + G\sqrt{r_1 r_2}}{1 - G\sqrt{r_1 r_2}} \right)^2 \xrightarrow{\Delta G < 2dB} G\sqrt{r_1 r_2} < 0.17 \xrightarrow{G = 30dB} \sqrt{r_1 r_2} < 1.7 \times 10^{-4}$$

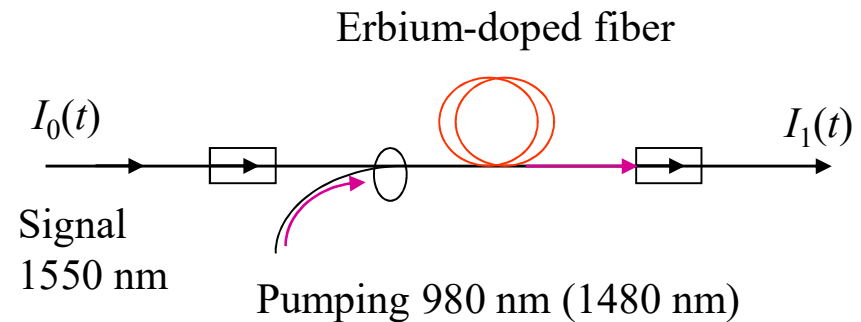
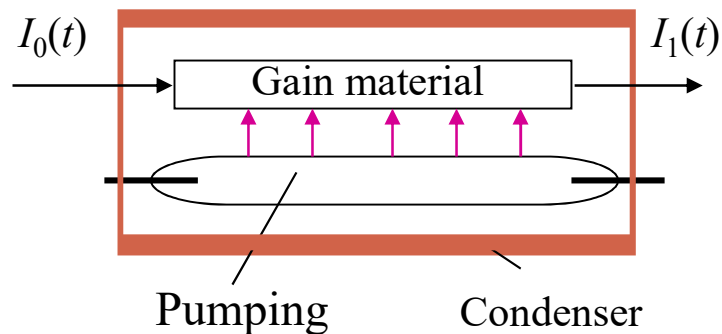
r_1 and $r_2 \approx 0$, regenerative amplifier $\Rightarrow$ traveling wave amplifier $G = G_s$

❖ Classification by functionality

- Preamplifier: weak signal working regime, small noise coefficient
- Power amplifier: strong signal, gain saturated state, large saturated output power
- Line amplifier: loss compensation of fiber in the system

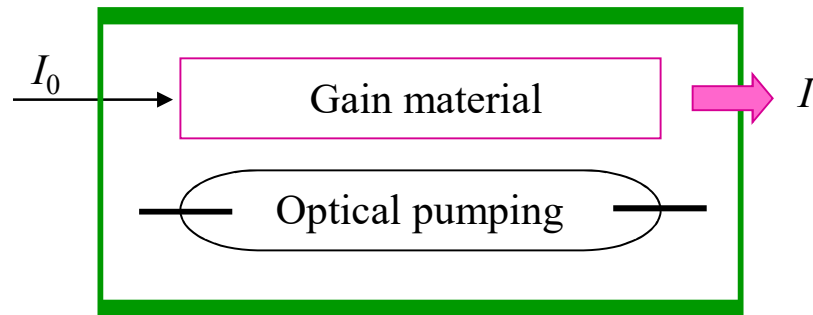
❖ Classification by optical excitation

- Uniform Transverse excitation: Δn , g^0 , and I_s are all constant (solid-state or semiconductor amplifier)
- Non-Uniform Longitudinal excitation: Δn , g^0 , and I_s are related to transmission distance (fiber amplifier)



5.2 Gain Characteristics of the Uniform Excited Continuous Laser Amplifier

- (1) Gain $\begin{cases} \text{weak signal gain} \\ \text{Saturated gain} \end{cases}$ (2) Maximum output intensity (saturated output power) (3) Gain spectrum



Assume: homogeneous broadening, average loss coefficient α , central frequency ν_0

Net gain coefficient
$$\frac{dI(z)}{I(z)dz} = g_m \left[1 + \frac{I(z)}{I_s} \right]^{-1} - \alpha$$

❖ Gain of the laser amplifier for weak signal

Weak signal $I(z) \ll I_s$

$$G^0(\nu_0) = \frac{I(l)}{I_0} = \exp[(g_m - \alpha)l] \quad \text{preamplifier}$$

❖ Gain of the laser amplifier for strong signal (saturated)

The incident signal is strong or the gain material is long enough $I(z) \sim I_s$



$$\frac{dI(z)}{I(z)dz} = g_m \left[1 + \frac{I(z)}{I_s} \right]^{-1} - \alpha$$

$$g_m dz = \frac{\left[1 + I(z)/I_s \right] dI(z)}{I(z) \left\{ 1 - \left[1 + I(z)/I_s \right] \alpha / g_m \right\}}$$

Integrate within the total length of the amplifier

Output intensity $\ln \left[\frac{I(l)}{I_0} \right] = (g_m - \alpha)l + \frac{g_m}{\alpha} \ln \frac{g_m - \alpha [1 + I(l)/I_s]}{g_m - \alpha [1 + I_0/I_s]}$

Gain of the Amplifier $\ln \left[\frac{G(\nu_0)}{G^0(\nu_0)} \right] = \frac{g_m}{\alpha} \ln \frac{g_m - \alpha [1 + I(l)/I_s]}{g_m - \alpha [1 + I_0/I_s]}$

I_0 is known, amplifier's g_m , l , and α can be measured, I_s is known

❖ Maximum output intensity

Due to gain saturation, when the input signal is strong or the gain material is long enough, amplifier gain coefficient will decrease, net gain coefficient = 0, intensity will no longer increase.

$$\frac{dI(z)}{I(z)dz} = g_m \left[1 + \frac{I(z)}{I_s} \right]^{-1} - \alpha = 0 \quad \Rightarrow \quad I_m = I_s \left(\frac{g_m}{\alpha} - 1 \right)$$

❖ Width of amplifier gain spectrum and output line profile

- Small signal homogeneous broadening optical amplifier $\delta\nu = \Delta\nu_H \sqrt{\frac{\ln 2}{\ln G^0(\nu_0) - \ln 2}}$

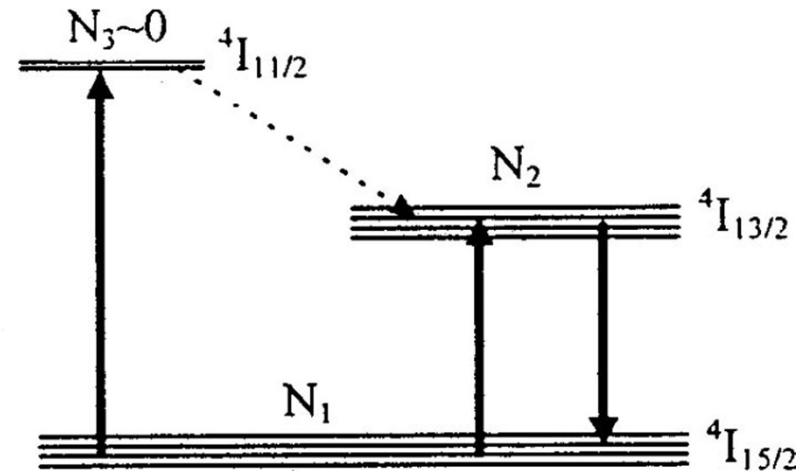
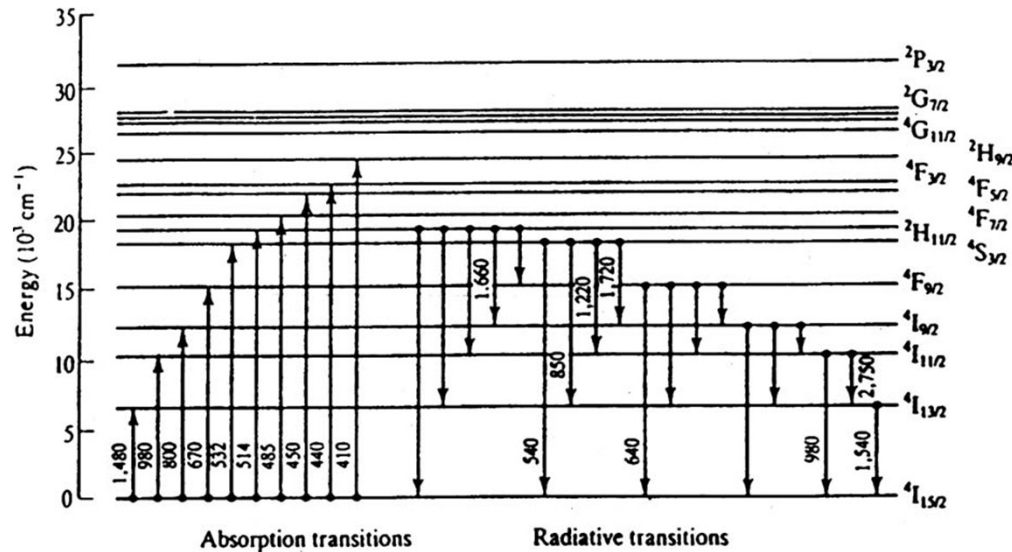
when $G^0(\nu_0) > 4$ $\delta\nu < \Delta\nu_H$ $G^0(\nu_0) \uparrow \Rightarrow \delta\nu \downarrow$

- Strong signal homogeneous broadening optical amplifier
- ✓ Saturation effect at central frequency is strong
 - ✓ Other frequencies saturation is weak

$$I(l) \uparrow \Rightarrow \delta\nu \uparrow \Rightarrow \delta\nu > \Delta\nu_H$$

5.3 Gain Characteristics of the Longitudinal Excited Continuous Laser Amplifier

Erbium-Doped Fiber Amplifier: EDFA



- Why EDFA belongs to continuous laser amplifier?
- ✓ Optical signal transition rate: $10^8 \sim 10^{10}$ b/s ($T = 10^{-8} \sim 10^{-10}$ s), erbium ions: $T_1 = 10^{-2}$ s, very short action time, Δn only has small fluctuation near the steady-state value
- Characteristics of longitudinal optical excitation

$$I(z), I_p(z) \Rightarrow g^0(z), \Delta n^0(z), I_s(z)$$